

Erdős Problem #872

EP-872

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Consider the two-player game in which players alternately choose integers from $\{2, 3, \dots, n\}$ to be included in some set A (the same set for both players) such that no $a \mid b$ for $a, b \in A$.

The game ends when no legal move is possible. One player wants the game to last as long as possible, the other wants the game to end quickly. How long can the game be guaranteed to last for?

At least ϵn moves? (For $\epsilon > 0$ and n sufficiently large.) At least $(1 - \epsilon)\frac{n}{2}$ moves?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>